

The Greenhouse Effect

Think in one chain: incoming solar radiation → reflection/absorption → outgoing infrared → equilibrium.

THE RADIATION PICTURE

Sun

- Photosphere $\approx 5800\text{ K}$ → near-black-body spectrum.
- Peak is in the visible range by Wien's law.
- Solar radiation spans UV, visible and infrared.

Earth

- Much cooler than the Sun → re-emits mainly longwave infrared.
- Atmosphere is relatively transparent to incoming shortwave sunlight.
- Greenhouse gases absorb much of outgoing IR and re-emit it in all directions.

MUST-KNOW FORMULAS

$$P_{\odot} = \sigma A_{\odot} T_{\odot}^4 \cdot I = P_{\odot} / (4\pi d^2)$$

$$P_{\text{in}} = S \pi r^2 (1 - \alpha) \cdot P_{\text{out}} = \epsilon \sigma (4\pi r^2) T^4$$

$$\text{At equilibrium: } P_{\text{in}} = P_{\text{out}}$$

Solar constant: $S \approx 1361\text{ W m}^{-2}$. Earth's average albedo: $\alpha \approx 0.3$.

ALBEDO & EMISSIVITY

Albedo α

- Fraction of incoming radiation reflected.
- $\alpha = 0$ → all absorbed; $\alpha = 1$ → all reflected.
- Lower α → more absorbed solar power → warming pressure.

Emissivity ϵ

- How effectively a real surface emits compared with a black body at the same T.
- Black body: $\epsilon = 1$.
- Lower emissivity → less outgoing radiation at the same T.

NO ATMOSPHERE vs WITH ATMOSPHERE

No atmosphere

- Set absorbed solar power = emitted thermal power.
- With $S = 1361\text{ W m}^{-2}$ and $\alpha = 0.30$ → equilibrium $T \approx 255\text{ K}$ ($-18\text{ }^{\circ}\text{C}$).

With atmosphere

- Greenhouse gases absorb outgoing longwave IR and re-emit it, including downward.
- Natural greenhouse effect raises average surface temperature to about 288 K ($15\text{ }^{\circ}\text{C}$).

CAUSES OF WARMING

What shifts the equilibrium?

- Solar constant $\uparrow$ → $P_{\text{in}} \uparrow$.
- Albedo $\downarrow$ → less reflection → $P_{\text{in}} \uparrow$.
- Emissivity $\downarrow$ → $P_{\text{out}} \downarrow$.
- Enhanced greenhouse effect → more outgoing IR absorbed/re-emitted downward → net $P_{\text{out}} \downarrow$.
- Ice-albedo feedback: ice melts → α falls → more energy absorbed → more warming → more melting.

MITIGATION MEMORY

Main strategies in the material

- Greater efficiency · coal/oil → natural gas · combined heat & power · renewables & nuclear · electric vehicles · carbon capture & storage.
- Memory cue: reduce energy waste + reduce carbon emissions + remove/store CO_2 .

STUDY METHOD: cover the right side / formula boxes → recall from memory → check → repeat.